

Vasanth Bhaskara

Data Science at ZS | NIT Calicut

[LeetCode Profile](#), [Portfolio Website](#)

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EDUCATION

B.Tech. in Mechanical Engineering

Dec 2020 – May 2024

National Institute of Technology (NIT), Calicut

- Cumulative GPA: **9.25/10.0**, Department **Rank 3**
- Relevant Coursework: Machine Learning, Statistics, Probability, Computer Programming

EXPERIENCE

Decision Analytics Associate

ZS Associates, Pune

Data Science and Advanced Analytics for Commercial Applications

Feb 2025 – Present

- Developed **promotion impact models** for **Johnson & Johnson's** immunology brands using **Linear** and **Bayesian regression** to quantify channel-level contribution and optimize omnichannel marketing effectiveness.
- Engineered **ETL** and analytical **data pipelines** using **SQL**, **AWS Redshift**, and **Pandas** for large-scale promotional and sales datasets following **medallion-style** data architecture principles.
- Built **feature engineering** and transformation pipelines including ad stocking, **normalization**, lagging, and logarithmic transforms to model **carryover effects** and improve **attribution** accuracy across marketing channels

Condition Monitoring Engineer

Hindustan Petroleum (HP), Visakhapatnam

Engineering for Large-Scale Critical Equipment Control and Monitoring

Aug 2024 – Jan 2025

- Conducted equipment monitoring using machine health monitoring software like Honeywell Forge and Bently Nevada.

SKILLS

- **Tech Stack:** Python, SQL, Databricks, Git, Streamlit, Docker, Excel
- **Frameworks:** PyTorch, Pandas, PySpark, LangChain, HuggingFace, FastAPI

PROJECTS

Biomedical Retrieval-Augmented Generation (RAG) System

May 2026

Link: [GitHub](#)

- Built a biomedical **RAG** pipeline over **PubMed** abstracts using **TF-IDF retrieval**, **recursive chunking**, LLM-based grounded generation, **FastAPI** serving, and **Docker-based containerized** deployment for medical question answering.
- Evaluated retrieval quality and response grounding using **RAGAS** metrics and chunking ablation experiments, while optimizing deployment efficiency by replacing heavyweight **Sentence Transformer** + **FAISS** retrieval with lightweight lexical search.

LLM Style Transfer via Parameter-Efficient Fine-Tuning

Feb 2026

Link: [GitHub](#)

- Fine-tuned **LiquidAI LFM2-1.2B** using **LoRA-based parameter-efficient fine-tuning** on **2,048** style-transfer examples to control generation style while training **<0.05%** of model parameters.
- Evaluated outputs using **LLM-as-a-Judge methodology** using **Gemini-2.5-Flash**, and **negative Log likelihood** metrics.
- Built **PyTorch** + **HuggingFace** pipelines for training, fine-tuning, and evaluation.

LSTM Based Real-time Music Generation Application

Jan 2026

Link: [Live Demo](#), [GitHub](#)

- Trained a character-level **LSTM** on ~800 songs with **embedding**, **sequence batching**, and **multinomial sampling** with **temperature control** to learn **next-token probability** distributions for symbolic music generation.
- Deployed a **Streamlit**-based interactive **inference** system using pretrained weights with configurable generation parameters, real-time audio synthesis, and live web deployment.

COURSES

- **Verified Certifications:** Databricks Certified - Data Engineer Associate, [Link](#)
- **Online Courses:** Stanford Online - Deep Learning Certificate, [Link](#)